# Pandas Data Analysis Workflow

*Prai sob dataset e kaj lage emon step-by-step guide. Protita method er pase comment diye bojhano hoyeche.*

## Workflow ek nojore

**Load → Inspect → Clean → Filter → Transform → Group/Summary → Merge → Outlier → Visualize → Save**

Ei order follow korle prai jekono dataset e kaj kora jay.

## 1. Setup ar Data Load

```
import pandas as pd      # data niye kaj korar main library
import numpy as np       # number/math er jonno

df = pd.read_csv("data.csv")        # CSV file porte hoy
# df = pd.read_excel("data.xlsx")   # Excel file hole eta use korbe
```

| Method | Ki kaj kore |
|---|---|
| `pd.read_csv()` | CSV file theke data niye DataFrame (table) banay |
| `pd.read_excel()` | Excel file theke data pore |

## 2. Data ke Chena (Inspect)

Kaj shuru korar age dekhte hobe data ta kemon.

```
df.shape             # (row, column) koyta ache
df.head()            # prothom 5 ta row dekhay
df.info()            # column type ar null koyta ta dekhay
df.describe()        # number column er min, max, average ityadi
df.columns.tolist()  # sob column er nam list akare
df.dtypes            # protita column er data type
```

| Method | Ki kaj kore |
|---|---|
| `shape` | Moot row ar column er songkha bole |

| Method | Ki kaj kore |
|---|---|
| `head()` | Prothom kotok row dekhay, data er dhoron bujhte |
| `info()` | Column type, null nai ki ache, memory use |
| `describe()` | Statistics summary: mean, min, max, quartile |
| `columns` | Column er nam gulo dey |
| `dtypes` | Kon column number, kon text, kon date ta bole |

# 3. Cleaning (Data porishkar kora)

## Column nam thik kora

```
df.columns = df.columns.str.strip().str.lower().str.replace(" ", "_")
# strip()   -> age pichoner faka space muche
# lower()   -> sob choto hater likhe
# replace() -> space er jaygay underscore boshay
```

## Missing value (khali ghor) handle kora

```
df.isna().sum()                                    # kon column e koto ta
khali
df = df.dropna(subset=["important_col"])           # dorkari column khali
hole row bad
df["age"] = df["age"].fillna(df["age"].median())   # number: median diye
bhorai
df["city"] = df["city"].fillna("Unknown")          # text: default likhe
bhorai
```

| Method | Ki kaj kore |
|---|---|
| `isna().sum()` | Protita column e koyta missing value ache gune |
| `dropna()` | Missing value thaka row fele dey |
| `fillna()` | Missing jaygay notun value boshay |
| `median()` | Moddhoman ber kore (outlier thakle mean er cheye bhalo) |

### Duplicate muche fela

```
df.duplicated().sum()        # koyta duplicate row ache
df = df.drop_duplicates()    # duplicate row gulo fele dey
```

### Data type thik kora

```
df["date"] = pd.to_datetime(df["date"], errors="coerce")   # text ke date
banay
df["price"] = pd.to_numeric(df["price"], errors="coerce")  # text ke number
banay
df["category"] = df["category"].astype("category")          # memory kom,
speed beshi
# errors="coerce" -> convert na hole error na diye NaN (khali) kore dey
```

### Text clean

```
df["name"] = df["name"].str.strip().str.title()   # space muche, Prothom
Okkhor Boro hater
```

## 4. Filter ar Select

```
df[["name", "price"]]                              # sudhu ei duto column nao
df[df["price"] > 500]                              # price 500 er beshi row
df[(df["price"] > 500) & (df["city"] == "Dhaka")]  # duto shorto ekshathe (&
= AND)
df[df["city"].isin(["Dhaka", "Chattogram"])]       # list er moddhe thaka
value
df.query("price > 500 and city == 'Dhaka'")        # same kaj, porte sohoj
```

| Method / Symbol | Ki kaj kore |
|---|---|
| `df[["a","b"]]` | Nirdishto column bachai kore |
| `df[condition]` | Shorto milay emon row dekhay |
| `&` | AND (duto shorto-i sotti hote hobe) |
| `or (pipe)` | OR (je kono ekta sotti hole hobe) |
| `isin()` | Value ti list er moddhe ache kina check kore |
| `query()` | Shorto string akare likhe filter kore |

# 5. Notun Column Banano

```
df["total"] = df["price"] * df["qty"]      # duto column gun kore notun
column
df["year"] = df["date"].dt.year            # date theke sal ber kora
df["month"] = df["date"].dt.month          # date theke mash ber kora
df["price_group"] = pd.cut(
    df["price"],
    bins=[0, 100, 500, 1000, np.inf],      # range gulo
    labels=["low", "mid", "high", "premium"]  # protita range er nam
)
```

| Method | Ki kaj kore |
|---|---|
| `df["new"] = ...` | Hishab kore notun column joge |
| `.dt.year / .dt.month` | Date column theke sal, mash alada kore |
| `pd.cut()` | Number ke range onujayi group e bhag kore |

# 6. Group ar Summary (sobcheye dorkari)

```
df.groupby("city")["total"].sum()    # city onujayi total sales

df.groupby("city").agg(
    total_sales=("total", "sum"),    # moot bikri
    avg_price=("price", "mean"),     # gor dam
    orders=("total", "count"),       # koyta order
).reset_index().sort_values("total_sales", ascending=False)

pd.pivot_table(df, values="total", index="city", columns="month",
aggfunc="sum")
```

| Method | Ki kaj kore |
|---|---|
| `groupby()` | Ekoi value er row gulo ekshathe group kore |
| `agg()` | Ekshathe onek dhoroner hishab (sum, mean, count) kore |
| `reset_index()` | Group er nam ke abar normal column banay |
| `sort_values()` | Boro theke choto ba choto theke boro sajay |

| Method | Ki kaj kore |
|---|---|
| `pivot_table()` | Excel pivot er moto table banay (row x column) |
| `value_counts()` | Ekoi value koyobar ache gune |

## 7. Merge / Join (Table jora lagano)

```
merged = df.merge(other_df, on="customer_id", how="left")   # common column diye jora
combined = pd.concat([df1, df2], ignore_index=True)         # niche niche jora
```

| Method | Ki kaj kore |
|---|---|
| `merge()` | Common column (jemon id) diye duto table pashapashi jore (SQL join er moto) |
| `how="left"` | Bam table er sob row rakhe, dan table theke milaya nei |
| `concat()` | Ek table er niche arekta table jore dey |

## 8. Outlier Check (ashbhabik value khuje ber kora)

```
q1, q3 = df["price"].quantile([0.25, 0.75])   # 25% ar 75% position er value
iqr = q3 - q1                                  # duto er parthokko
outliers = df[(df["price"] < q1 - 1.5*iqr) | (df["price"] > q3 + 1.5*iqr)]
# ei range er baire jegulo, segulo outlier
```

## 9. Quick Visualization (chart)

```
df["price"].hist()                                      # value er distribution
df.groupby("month")["total"].sum().plot(kind="line")    # mash onujayi trend
df["city"].value_counts().plot(kind="bar")              # city wise count
```

| Chart | Kokhon use korbe |
|---|---|
| `hist()` | Ekta number column er spread dekhte |
| `line` | Somoy er sathe poriborton (trend) dekhte |

| Chart | Kokhon use korbe |
|---|---|
| `bar` | Category gulo tulona korte |

## 10. Save

```
df.to_csv("clean_data.csv", index=False)      # CSV hishebe save
df.to_excel("clean_data.xlsx", index=False)   # Excel hishebe save
# index=False -> extra index column save hobe na
```

## Cheat Sheet (Choto Mone Rakhar Jonno)

| Kaj | Method |
|---|---|
| Dekha | `head`, `info`, `describe` |
| Clean | `dropna`, `fillna`, `drop_duplicates`, `astype` |
| Filter | `df[condition]`, `query` |
| Notun column | `df["new"] = ...` |
| Summary | `groupby`, `pivot_table`, `value_counts` |
| Jora | `merge`, `concat` |
| Save | `to_csv`, `to_excel` |

**Tip:** Kono notun dataset pelei ei order e kaj shuru koro: Load → Inspect → Clean → Transform → Analyze → Save.